

Sarah E. Baum

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EDUCATION

Harvard University

2022–2027 (expected)

Ph.D. in Population Health Sciences (Department of Global Health and Population)

Fields: Health Economics, Infectious Disease Epidemiology

Dissertation: Allocation of infectious disease screening and its implications for disease surveillance

Committee: Nicolas Menzies (Chair), Jessica Cohen, Leila Agha

Barnard College, Columbia University

2013–2017

B.A. in Economics with honors, magna cum laude

AREAS OF EXPERTISE

Health Economics, Health Systems, Causal Inference, Impact and Policy Evaluation, Infectious Disease Epidemiology, Public Health, Global Health and Development.

GRANTS, FELLOWSHIPS, AND AWARDS

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| 2026 | Social Policy Research Grant, Horowitz Foundation for Social Policy |
| 2023–2026 | NIH National Institute of Allergy and Infectious Diseases T32 Award |
| 2018–2019 | Princeton in Africa Fellowship |

PUBLICATIONS

1. **Baum, S.E.**, Agha, L., Menzies, N.A., and Cohen, J. (2026) Prenatal syphilis screening mandates and maternal syphilis case detection. *JAMA Health Forum*, 7(3): e260123.
2. **Baum, S.E.**, Pelissari, D.M., Dockhorn Costa, F., Harada, L.O., Sanchez, M., Bartholomay, P., Cohen, T., Castro, M.C., and Menzies, N.A. (2024) Surveillance for TB drug resistance using routine rapid diagnostic testing data: Methodological development and application in Brazil. *PLOS Computational Biology*, 20(12): e1012662.
3. Graham, S.B., Machalaba, C., **Baum, S.E.**, Raufman, J., and Hill, S.E. (2023) Applying a One Health lens to understanding the impact of climate and environmental change on healthcare-associated infections. *Antimicrobial Stewardship & Healthcare Epidemiology*, 3(1): e93.
4. **Baum, S.E.** and Benshaul-Tolonen, A. (2021) Extractive industries and gender equality. *Review of Environmental Economics and Policy*, 15(2): 195–215.
5. **Baum, S.E.**, Reddy, V., Vora, N.M., Balter, S., Daskalakis, D., Barbot, O., Misener, M., Rakeman, J., Rojas, J., Starr, D., Waechter, H., Zucker, J., and Lee, D. (2020) Cost comparison between 2 responses to hepatitis A virus incidents in restaurant food handlers—New York City, 2015 and 2017. *Journal of Public Health Management and Practice*, 26(2): 176–179.
6. **Baum, S.E.**, Machalaba, C., Daszak, P., Salerno, R.H., and Karesh, W.B. (2017) Evaluating One Health: Are we demonstrating effectiveness? *One Health*, 3: 5–10.
7. Machalaba, C., Romanelli, C., Stoett, P., **Baum, S.E.**, Bouley, T.A., Daszak, P., and Karesh, W.B. (2015) Climate change and health: Transcending silos to find solutions. *Annals of Global Health*, 81(3): 445–458.

WORKING PAPERS AND RESEARCH IN PROGRESS

1. Selection and Surveillance: Linking Shocks to Safety Net Health Care Access and STI Case Detection

2. Physician Responses to Changes in Local Epidemiology: A Movers Design Leveraging Shocks to Lyme Disease Risk
3. Identifying Pathways to TB Drug Resistance in Brazil *with Nicolas Menzies*

INVITED PRESENTATIONS

- 2026 “Quasi-experimental Approaches to Program and Policy Evaluation.” **San Diego State University, DrPH Program, PH 910: Research Methods & Ethics.** (Guest Lecture)
- 2026 “Selection and Surveillance: The Impact of Abortion Funding Cuts on Safety Net Access and STI Detection.” **ASHEcon Annual Conference**, Minneapolis, MN.
- 2026 “Legal Mandates, Provider Screening, and Syphilis Detection in Pregnancy.” **AcademyHealth Annual Research Meeting**, Seattle, WA. (Poster Presentation)
- 2026 “Shocks to the U.S. Safety Net and STI Detection.” **Department of Global Health and Population Research Seminar, Harvard University.**
- 2025 “Surveillance for TB Drug Resistance Using Routine Rapid Diagnostic Testing Data.” **Department of Global Health and Population Research Seminar, Harvard University.**
- 2022 “Strengthening One Health and Veterinary Programs for Global Health Security in Africa.” **Global Health Security Conference**, Singapore.
- 2018 “The Natural Resource Curse and Women.” **World Bank Land and Poverty Conference**, Washington, DC. (Poster Presentation)
- 2016 “Beyond Operationalizing: The Need for Evaluation in One Health.” **Consortium of Universities in Global Health Conference**, San Francisco, CA. (Poster Presentation)

TEACHING EXPERIENCE

- 2026 **Introduction to Economics with Applications to Health and Development**
Teaching Fellow for James Potter (Master’s students)
- 2026 **Econometric Methods in Impact Evaluation**
Teaching Fellow for Jessica Cohen (Ph.D. students)
- 2025 **Introduction to Economics with Applications to Health and Development**
Teaching Fellow for James Potter (Master’s students)
- 2025 **Econometric Methods for Health Policy**
Teaching Fellow for Nidhiya Menon (Master’s and Ph.D. students)
- 2024 **Introduction to Economics with Applications to Health and Development**
Teaching Fellow for Margaret McConnell (Master’s students)

RESEARCH AND PROFESSIONAL EXPERIENCE

Harvard University

Doctoral Researcher

Boston, MA

August 2022 – Present

- Estimated impact of shocks to safety-net providers on chlamydia testing, case detection, and population health outcomes using quasi-experimental design and mathematical modeling
- Led quasi-experimental evaluation of the effect of state-level syphilis screening mandates on prenatal syphilis detection using national vital statistics data and hospital discharge claims
- Developed a statistical approach to improve surveillance of drug-resistant tuberculosis using national notification registry data, in collaboration with Brazil’s Ministry of Health

Mass General Brigham

Consultant

Boston, MA

July 2025 – February 2026

- Identified potential public and commercial administrative and EHR data sources for quasi-experimental evaluation of antenatal steroid exposure

EcoHealth Alliance

Health & Policy Research Assistant

New York, NY

July 2021 – July 2022

- Assessed multi-sectoral strategies to strengthen health security across ten African countries
- Analyzed pandemic preparedness and antimicrobial resistance data for World Bank partners
- Conducted economic evaluation of interventions against Rift Valley fever in South Africa

Abdul Latif Jameel Poverty Action Lab (J-PAL)

Senior Policy Associate

Boston, MA

August 2019 – July 2021

- Synthesized lessons from randomized controlled trials of environment, energy, and climate change programs and conducted policy outreach to policymakers and NGOs
- Cultivated new research partnerships between academics, NGOs, and governments to design, pilot, evaluate, and scale effective poverty alleviation policies and programs
- Worked with PIs to design research agenda for \$25 million climate research and policy funding initiative and managed semiannual funding competitions
- Collaborated with senior leadership on increasing diversity of J-PAL's network of 200+ junior researchers

Population Services International

Research, Monitoring & Evaluation Fellow

Johannesburg, South Africa

November 2018 – July 2019

- Led data reporting, audits, and data quality trainings across six countries for the HIV Self-Testing Africa (STAR) Initiative
- Facilitated DHIS2 training, piloting, and roll-out among voluntary medical male circumcision mobilizers in South Africa
- Developed and reviewed household survey instruments related to malaria and sexual and reproductive health
- Authored technical reports and publications, and managed inputs for donor deliverables in collaboration with leadership and country offices

Barnard College

Research Assistant

New York, NY

May 2017 – July 2019

- Co-authored a literature review and descriptive analysis on the gender impacts of extractive industries
- Assessed links between resource governance and gender equality for the Natural Resource Governance Institute

New York City Department of Health

Consultant

New York, NY

July 2017 – July 2018

- Led analysis comparing different responses to hepatitis A outbreaks
- Synthesized use cases for Regional Health Information Organizations in disease surveillance activities informed by qualitative focus groups with public health experts

United Nations Development Programme

Health & Innovative Financing Intern, HIV, Health and Development Group

New York, NY

June 2017 – November 2017

- Wrote briefs to engage non-health ministries in multi-sectoral responses to non-communicable diseases and collaborated on drafting a guidance note for the Global Fund's tuberculosis programming

TECHNICAL SKILLS

Software: R, Stata, ArcGIS, L^AT_EX

Languages: English (native), French (conversational)

REFERENCES

Nicolas A. Menzies, Ph.D.

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Department of Global Health and Population
Harvard T.H. Chan School of Public Health
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Leila Agha, Ph.D.

Associate Professor of Health Care Policy
Department of Health Care Policy
Harvard Medical School
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Jessica Cohen, Ph.D.

Professor of Health Economics
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